STAKE DXF

USER GUIDE · v1.22

Install · Usage · Help
Trimble TSC5 field controller app
TRIO Engineering

Companion to StakeDXF_UI_Slide_Deck.pdf

1. INSTALLATION (TRIMBLE TSC5)

StakeDXF ships as an Android APK for the Trimble TSC5.

File: dist/StakeDXF vX.Y.Z.apk (~65 MB)
Package: com.stakedxf.stakedxf

Steps
1. Copy the APK onto the TSC5 (USB File Transfer or your usual share).
2. Open Files / Downloads on the TSC5 and tap the APK.
3. If blocked, enable Allow from this source for that app.
4. Tap Install, then open StakeDXF from the app drawer.

If Install is greyed out
Company MDM may block unknown APKs. Ask IT for an unknown-source exception or push the APK through managed distribution.

Common mistakes
• Trying to open the APK on iPhone (Android only).
• Opening a zip/repo instead of the .apk.
• USB stuck on Charging only — switch to File Transfer.

2. USAGE — CONVERT (DWG → DXF)

Purpose: turn a Civil 3D drawing into a Trimble-stakeable DXF on the controller.

1. Open StakeDXF → CONVERT.
2. Tap the drawing slot and pick the office DWG / DXF.
3. Tap RUN CONVERT.
4. Confirm the stakeable entity count and proxy explode count.
5. Review the LAYERS list and tick which to include.
6. Tap SAVE DXF — writes into the TSC5 documents folder.
7. Move it into Trimble Data/Projects/<job>/.
8. Trimble Access: Map → Layer manager → Map files → selectable → Stakeout.

Pipeline (on-device)
- LibreDWG reads the DWG into DXF.
- ezdxf explodes ACAD_PROXY_ENTITY / AEC graphics into LINE / ARC / POLY.
- Non-stakeable entity types are filtered out.

Tip: office DWGs should be saved with proxy graphics preserved.

3. USAGE — PLOT (EXPORT POINTS)

Purpose: build a scaled staking sheet from selected points (and optional linework).

1. Export points from Trimble Access as CSV/TXT (PNEZD).
2. StakeDXF → PLOT.
3. Enter a JOB name.
4. IMPORT POINTS CSV / TXT.
5. Optionally LINK DXF LINEWORK and check the layers you want.
6. Set MARKER, LABEL, SCALE, and SHEET.
7. Optional: place library objects (hydrant / MH / sign).
8. Select which points appear on the sheet.
9. Tap CREATE STAKING PLOT PDF, or EXPORT CSV for a trimmed list.

Supported point formats
• PNEZD (headered or not)
• Common header aliases (Point #, Elev, Desc, …)

Linked DXF linework
• Reads LINE / LWPOLYLINE / POLYLINE / ARC / CIRCLE
• Layer checklist keeps only the linework you want on the plot

4. PLOT CUSTOMIZATION REFERENCE

Markers
Filled triangle · Triangle outline · Cross (+) · X · Large X · Circle · Dot · Large dot

Labels
Point number · Number + description · Number + elevation ·
Number + description + elevation · No labels

Layout
- Point list table off by default — more plan area for staking.
- Turn the table on for a control-note style coordinate list.
- Sheet: ANSI A–D, portrait/landscape, auto or fixed engineering scale.

Colors
- Full ACI 1–255, CTB-resolved swatches, and true-color HSV picker.

Examples
See dist/plot_examples/ for sample PDFs covering these combinations.

5. HELP & TROUBLESHOOTING

No stakeable entities after convert
• Confirm the DWG still contains proxy graphics (ACAD_PROXY_ENTITY).
• Civil objects saved without proxies cannot be exploded to linework.

Staking plot has no linework
• Link a DXF and enable Draw linked DXF linework.
• Ensure at least one layer is checked.
• Only LINE / LWPOLYLINE / POLYLINE / ARC / CIRCLE are drawn.

Points missing after import
• Use PNEZD or a headered CSV with Northing/Easting columns.
• Blank or malformed rows are skipped.

App will not install
• Must be the TSC5 (Android), not iPhone.
• Enable Allow from this source or ask IT for MDM approval.

6. QUICK REFERENCE

Home
  01 CONVERT   Recover Civil 3D linework
  02 PLOT   CSV + staking plot PDF

Convert
  Pick drawing → RUN CONVERT → tick layers → SAVE DXF → Trimble Map files

Plot
  Import CSV → (link DXF) → Marker / Label / Scale / Sheet →
  Select points → CREATE STAKING PLOT PDF   or   EXPORT CSV

Good defaults for staking
  Marker: Large X
  Label: Number + elevation
  Table: Off
  Layers: On (only what you need)

Runs on-device. No cloud. No tracking.